From Imbalance to Insight: Building Trustworthy and Explainable AI for ICU Risk Prediction

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Introduction and Background

PROBLEM
- ICU mortality prediction is critical but hindered by **severe class imbalance**
- Standard models optimize **accuracy**, missing rare **high-risk cases**

SOLUTION
- Evaluate **balance training** vs. standard approaches
- Build **interpretable, trustworthy models** for the minority-class detection

OBJECTIVE
- Determine which approach **(unbalanced vs. balanced)** better identifies high-risks ICU patients while maintaining clinical trustworthiness

Data	
Size	2250 rows x 44 columns
Target	bad_outcome
Source	eICU Research Database
Features	labs, vitals, demographics, clinical scores

Logistic Regression Baseline Model			
	Precision	**Recall**	**F1**
0	0.89	0.99	0.94
1	0.64	0.15	0.24
Macro Avg	0.77	0.57	0.59

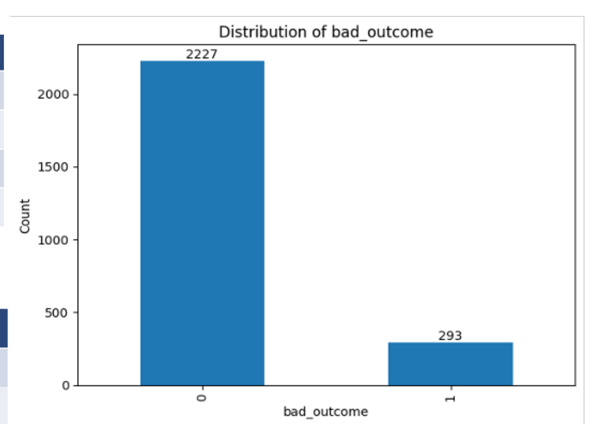


Fig. 1: Distribution of target variable, bad_outcome

Methodology

DATA + MODELS
- Merged **eICU records** into a single analysis dataset
- Evaluated **6 models** per scenario, plus a custom **stacked model**
- Base Learners: Logistic regression, XGBoost
- Meta-Learner: Logistic Regression

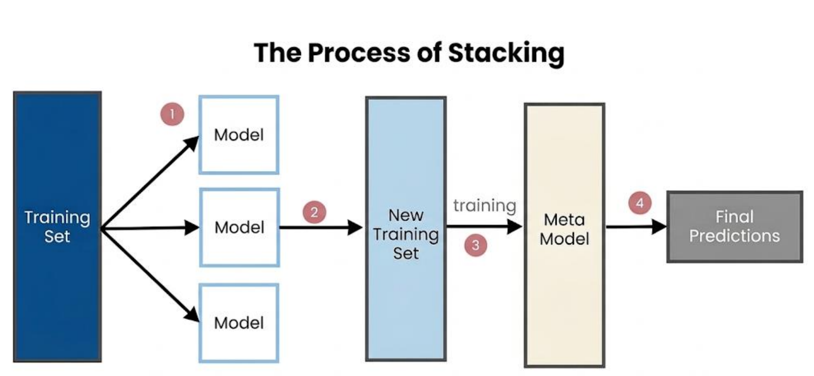


Fig. 2: Stacked model methodology

TRAINING
- **Stratified k-fold cross-validation** with 80/20 train-test split
- Default decision threshold = **0.5**

EVALUATION
- Prioritized: **Recall > Precision > F1**
- Used **PR-AUC** and **ROC-AUC** for overall performance

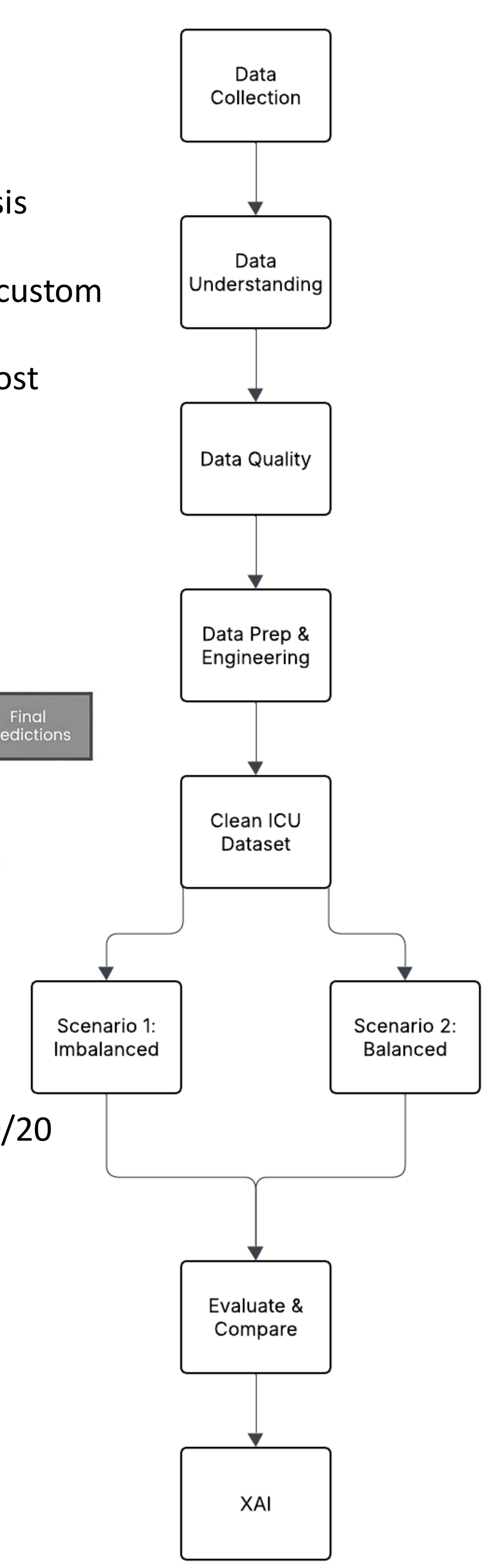


Fig. 3: Model Development Flowchart

Results

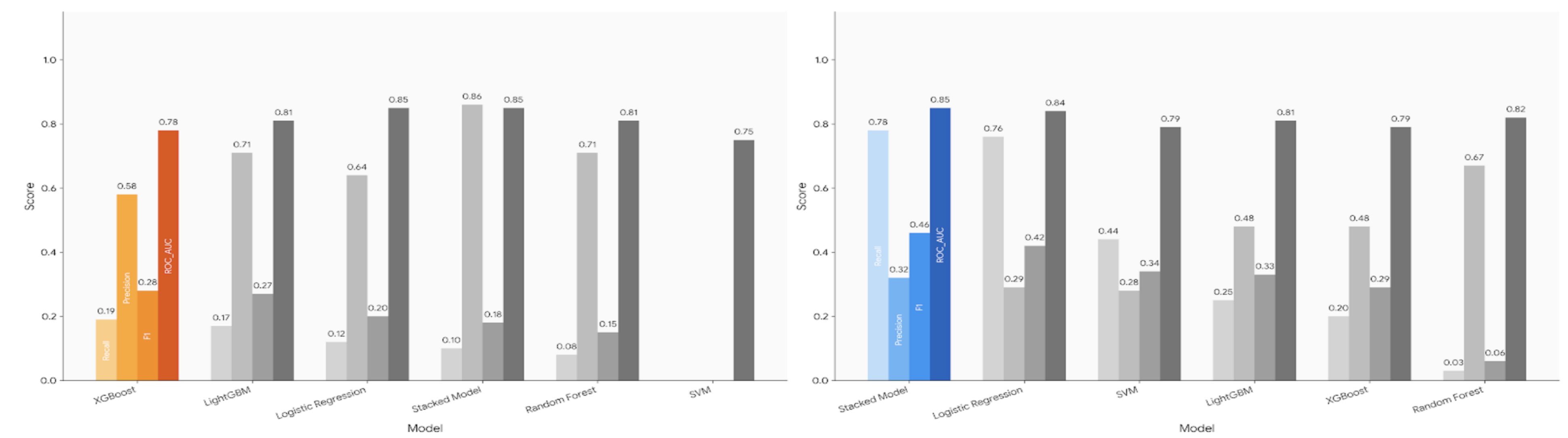


Fig. 4: Comparison of test metrics for all models trained on **Scenario 1: unbalanced (left)** vs. **Scenario 2: balanced (right)** distributions

XGBoost Model (Unbalanced) Classification Report			
	Precision	**Recall**	**F1**
0	0.90	0.98	0.94
1	0.58	0.19	0.28
Macro Avg	0.74	0.58	0.61

Stacked Model (Balanced) Classification Report			
	Precision	**Recall**	**F1**
0	0.96	0.78	0.86
1	0.32	0.78	0.46
Macro Avg	0.64	0.78	0.66

Table 1 and 2: Classification report comparison of **best-performing** models (XGBoost Unbalanced and Stacked Model Balanced)

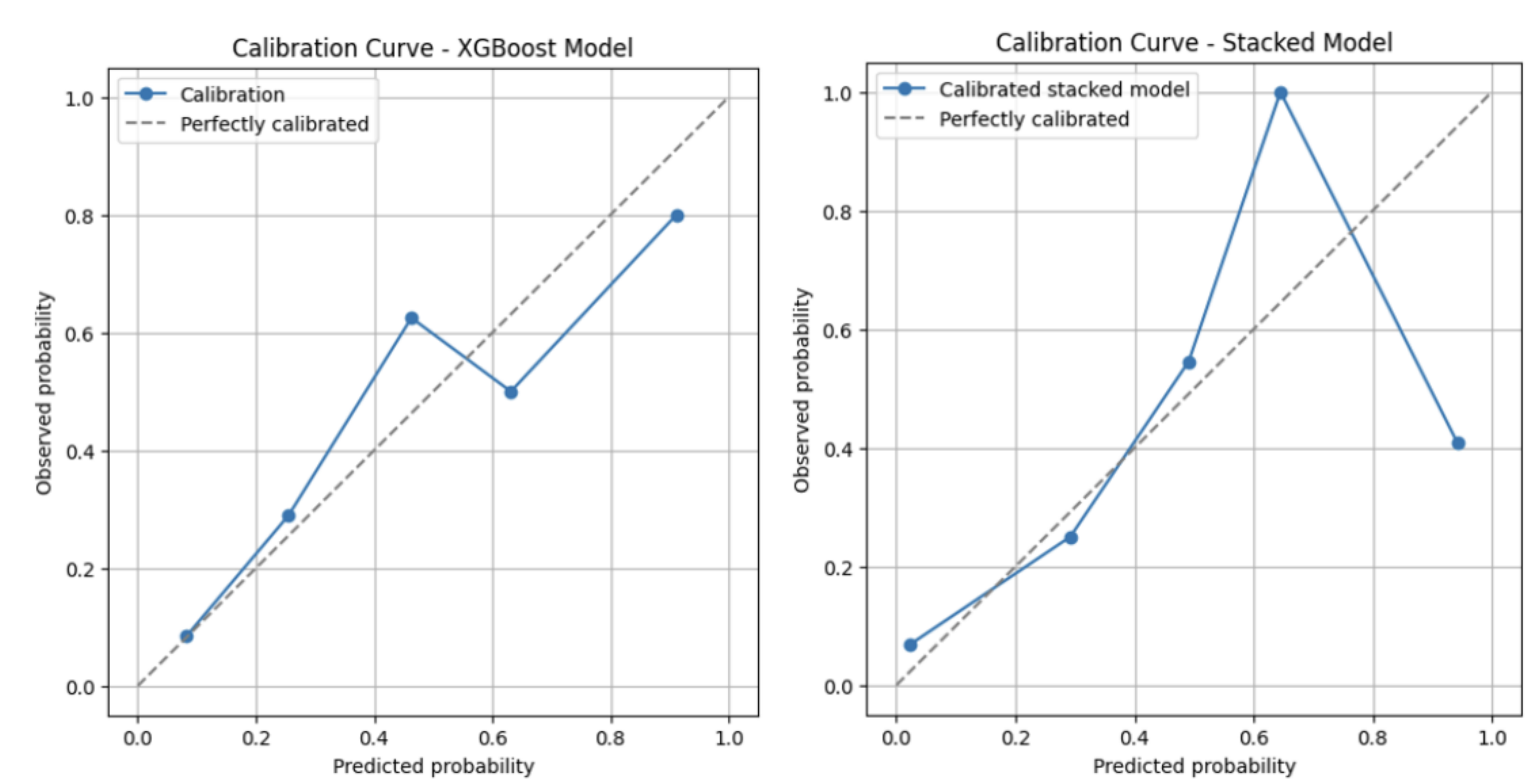


Fig. 5 and 6: Calibration curves between the different models, showing how confident the model is

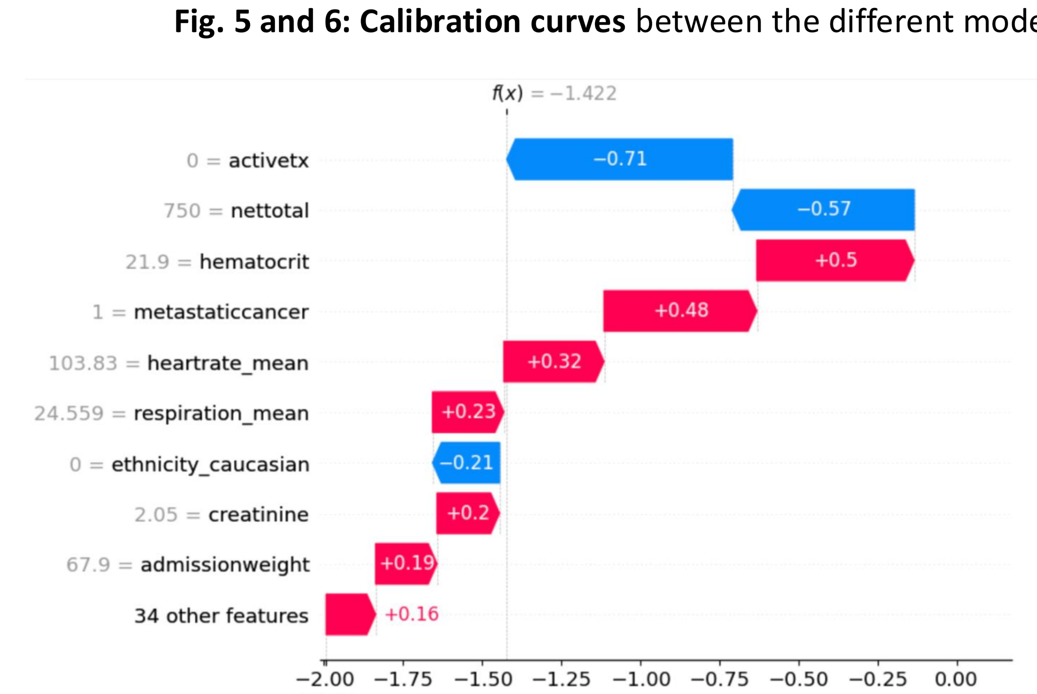


Fig 7: Local Interpretability using SHAP Values. Each horizontal bar represents the "push" a feature provides to move the model from the expected mean value $E[f(X)]$ to the specific predicted value $f(x)$.

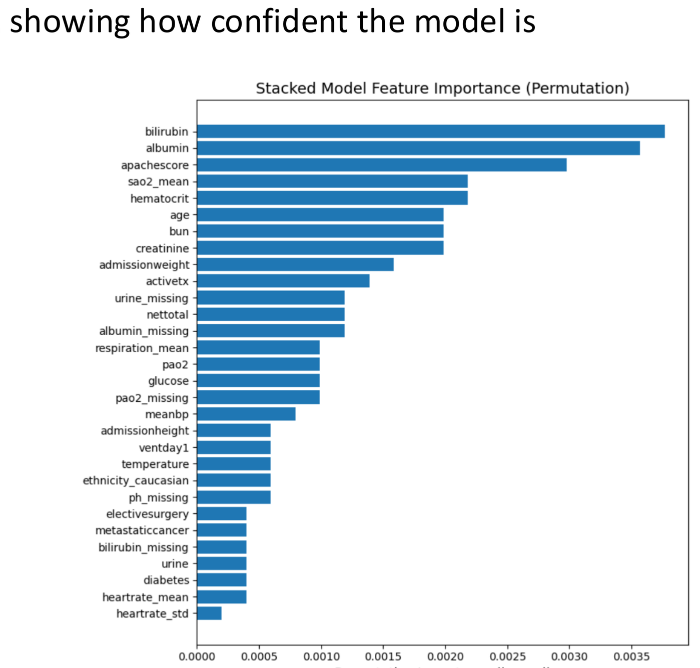


Fig 8: Permutation Feature Importance for the Stacked Ensemble Model. Features are ranked by the mean decrease in model performance when their values are randomly permuted.

Discussion

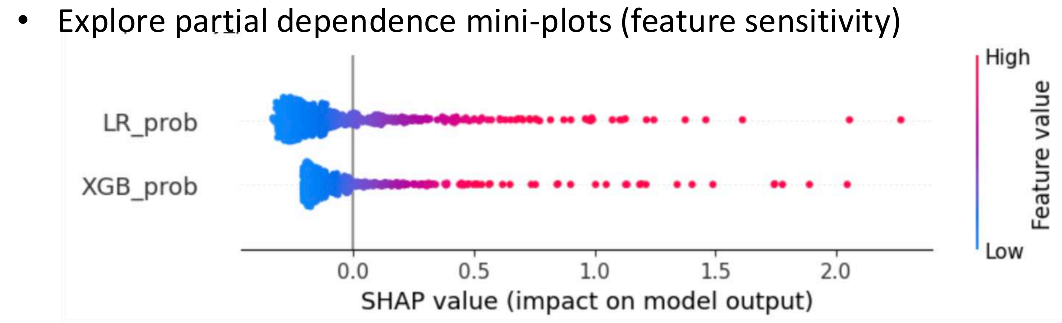
- Model performance must align with **clinical priorities**, not just accuracy
- **False negatives are critical** — missing high-risk patients is more dangerous than over-alerting
- Clear trade-off observed:
- **XGBoost** → higher recall (better at identifying high-risk patients)
- **Stacked model** → better balance (higher precision & specificity)
- **Class balancing strategies** improved learning of rare outcomes
- **Probability calibration** ensured risk % reflected real ICU prevalence
- **Isotonic calibration** improved probability alignment without fixed assumptions
- **Threshold tuning** allowed adjustment based on clinical needs
- Models can contextualize predictions by assigning each patient to a clinically interpretable risk band based on **calibrated model probabilities.**
- Effective deployment requires **discrimination + calibration + interpretability**, not just one metric

Fig 9: Risk Stratification Matrix for Clinical Decision Support. Model-derived probabilities are mapped onto calibrated risk bands to guide intervention. This calibration ensures that clinical resources are prioritized according to both the likelihood and the potential severity of the outcome.

Future Work

Patient ID	Bad Outcome Risk (%)	Reasons Why	Recommended Course of Action	Actual Outcome (Internal)
1234	0.5	age (↑), sao2_mean (↓), activetx(↑)	Standard Monitoring	0
4321	5.0	activetx (↓), age(↑), heartrate_mean(↑)	Monitor closely/early intervention	1
7890	0.0	age (↓), bun (↓), albumin (↓)	Standard Monitoring	0
4567	1.4	activetx (↓), meanbp (↑), heartrate_mean (↓)	Standard Monitoring	0
7654	13.2	sao2_mean (↓), apachescore (↑), heartrate_mean (↑)	Monitor closely/early intervention	1

- Deploy as a clinician-facing dashboard (Streamlit-style prototype)
- Include stacked risk + risk band + SHAP explanation
- Add comparison toggle (Stacked vs XGBoost)
- Add counterfactual "Model Second Opinion" widget
- Explore partial dependence mini-plots (feature sensitivity)



- The global SHAP analysis provides important insight into model behavior by highlighting how predicted probabilities influence overall predictions
- Higher feature values (red points) are associated with larger positive SHAP values, indicating increased predicted risk
- The agreement between SHAP values and predicted probabilities supports model interpretability and reliability
- Highlights the importance of probability-based features as key drivers of prediction outcomes

References

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